EXPERIMENT NO. 07

Aim: To implement monitoring using Prometheus / Grafana / Nagios

LO Mapping: LO5

LO5: Implement ML lifecycle management and automated workflows using MLflow, Airflow, and monitoring tools like Prometheus/Grafana/Nagios.

Theory:

Continuous monitoring is a process to detect, report, respond all the attacks which occur in its infrastructure. Once the application is deployed into the server, the role of continuous monitoring comes in to play. The entire process is all about taking care of the company's infrastructure and respond appropriately.

Why We Need Nagios tool?

Here, are the important reasons to use Nagios monitoring tool:

- Detects all types of network or server issues
- Helps you to find the root cause of the problem which allows you to get the permanent solution to the problem
- Active monitoring of your entire infrastructure and business processes
- Allows you to monitor and troubleshoot server performance issues
- Helps you to plan for infrastructure upgrades before outdated systems create failures
- You can maintain the security and availability of the service
- Automatically fix problems in a panic situation
- Nagios is the most popular, open source, powerful monitoring system for any kind of infrastructure. It enables organizations to identify and resolve IT infrastructure problems before they affect critical business processes. Nagios has the capability of monitoring applications, services and entire IT infrastructure.
- NRPE is known as Nagios Remote Plugin Executor. It is designed to execute plugins on remote systems. On Windows hosts, the equivalent remote agent is NSClient++, which listens for Nagios requests and executes local checks (disk, CPU, memory, services, etc.).
- NSClient++ runs as a Windows service on the remote host and waits for requests from the Nagios server (usually over the NRPE protocol, which NSClient++ also supports). When the Nagios server needs to check a resource on that host, it sends

a command; NSClient++ executes the check locally and returns the result to Nagios.

Note on running Nagios on Windows

Nagios Core is built for Linux and does not install natively on Windows. The standard practice for a Windows machine is to run the Nagios Core server inside a Docker container (using Docker Desktop with the WSL2 backend), and to monitor Windows hosts using NSClient++, which is the Windows counterpart of NRPE. All commands below are Windows PowerShell / Command Prompt commands.

1 - Pre-requisite: Install Docker Desktop

docker – - version
wsl - - install
wsl –set default-version 2

2 - Pull and Run the Nagios Core Docker Image

Pull a ready-made Nagios Core image and run it as a container, mapping the web port to the host.

docker pull jasonrivers/nagios:latest
docker run --name nagios4 -d -p 8080:80 jasonrivers/nagios:latest
docker ps

3 - Set the Apache/Nagios Admin Password

Set (or reset) the password for the nagiosadmin user inside the running container:

docker exec -it nagios4 htpasswd -c /opt/nagios/etc/htpasswd.users
nagiosadmin

4 - Access Nagios Web Interface

Access the Nagios web interface from a browser on the Windows host:

http://localhost:8080/nagios/
Prompting for Authentication Password -
username: nagiosadmin
Password: 123456 (which you entered while configuration)

5 - Install NSClient++ on the Windows Remote Host

On the Windows machine you want to monitor, download and silently install NSClient++ (the Windows equivalent of the NRPE agent):

https://github.com/mickem/nscp/releases/latest/download/NSCP-0.5.2-x64.msi" -OutFile "NSCP.msi"
Start-Process msiexec.exe -ArgumentList '/i NSCP.msi ADDLOCAL=ALL MODULES=CheckExternalScripts,NRPEServer /qn' -Wait

6 - Configure NSClient++

Edit the NSClient++ configuration file and add the Nagios server's IP address to the allowed hosts list:

NSClient++> notepad nsclient.ini
allowed hosts=127.0.0.1,192.168.1.100
Where 192.168.1.100 is your Nagios server (Docker host) IP address.
After saving the file, restart the NSClient++ service so the change takes effect:
Restart-Service nscp

7 - Verify Connectivity from Nagios

From the Nagios server (inside the Docker container), run the NRPE check against the Windows remote host to confirm connectivity. Here 192.168.1.50 is the remote Windows host's IP address.

docker exec -it nagios4 /opt/nagios/libexec/check_nrpe -H 192.168.1.50
NRPE v2.15

8 - Add the Windows Host in Nagios

Create a configuration file for the new host inside the Nagios container. We also define a service that continuously checks whether the host is reachable (PING check).

PS C:\Users\soham> docker exec -it nagios4 sh -c "vi /opt/nagios/etc/servers/MyWindowsHost001.cfg"
define host {
use windows-server
host_name Windows_Host_001

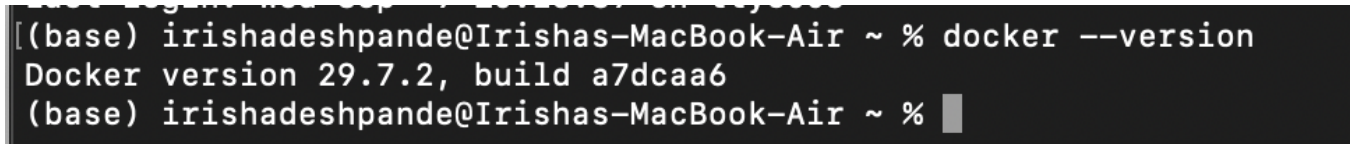
```
alias Windows Host 001
address 192.168.1.50
register 1
}
define service{
host_name Windows_Host_001
service_description PING
check_command check_ping!100.0,20%!500.0,60%max_check_attempts 2
check_interval 2
retry_interval 2
check_period 24x7
check_freshness 1
contact_groups admins
notification_interval 2
notification_period 24x7
notifications_enabled 1
register 1
}
```

Verify the configuration file for errors, then restart the Nagios container so the changes take effect:

9 - Check Host in Nagios Web Interface

Open the Nagios web interface (http://localhost:8080/nagios/) and check that the new Windows host has been added and is being monitored under Current Status -> Hosts.

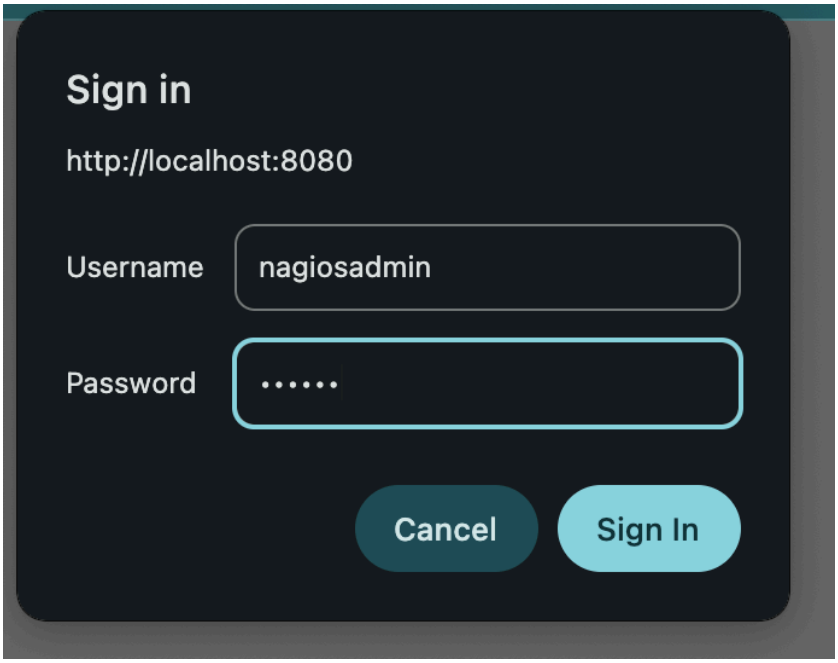
Implementation:



```
(base) irishadeshpande@Irishas-MacBook-Air ~ % docker pull jasonrivers/nagios:latest
latest: Pulling from jasonrivers/nagios
6fe99ef1bed3: Pulling fs layer
c42073aaf047: Pulling fs layer
254354b70ac0: Pulling fs layer
54b9e993e315: Pulling fs layer
80b6ab7aef5d: Pulling fs layer
a80db686d641: Pulling fs layer
6fe99ef1bed3: Pull complete
0d59a81a5608: Pull complete
7a7447982198: Pull complete
c5701b5e1540: Pull complete
1b9771f55a12: Pull complete
a03720c2c16b: Pull complete
e68fcda12c0c: Pull complete
f47c3af6bf44: Pull complete
cb259a83ac3d: Pull complete
fdbaa15b8986: Pull complete
748b632e75e7: Pull complete
9288c4992679: Pull complete
dc3a88012a3e: Pull complete
50ea270d34e9: Pull complete
37f75c8146e2: Pull complete
e4dc1d4e010c: Pull complete
```

```
(base) irishadeshpande@Irishas-MacBook-Air ~ % docker exec -it nagios4 htpasswd -c /opt/nagios/etc/htpasswd.users nagiosadmin
New password:
Re-type new password:
Adding password for user nagiosadmin

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug nagios4
    Learn more at https://docs.docker.com/go/debug-cli/
```



Nagios

Version **4.5.12** March 25, 2026

Process running with PID 22

Check for updates

A new version of Nagios Core is available!
Visit nagios.org to download Nagios 4.5.14.

General

- Home
- Documentation

Current Status

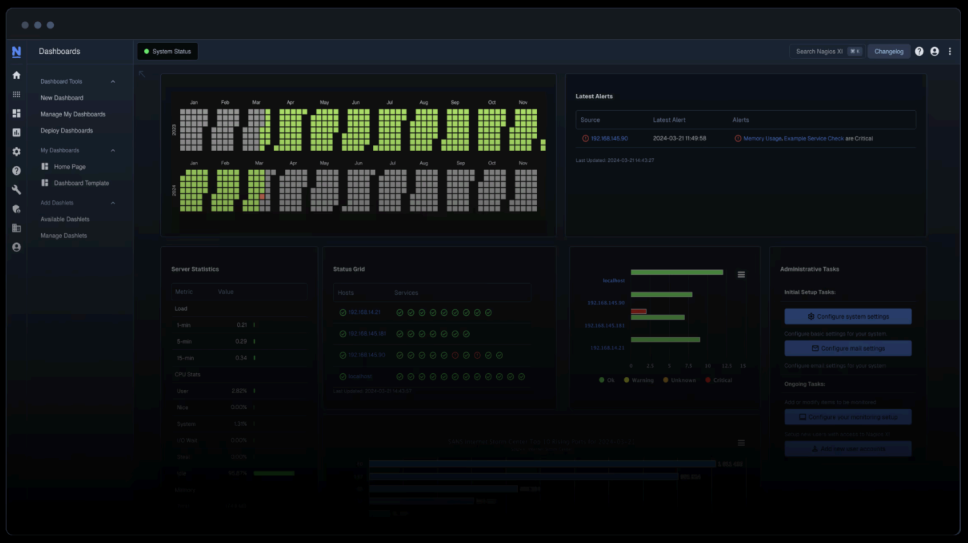
- Tactical Overview
- Map
- Hosts
- Services
- Host Groups
  - Summary
  - Grid
- Service Groups
  - Summary
  - Grid
- Problems
  - Services (Unhandled)
  - Hosts (Unhandled)
  - Network Outages

Quick Search:

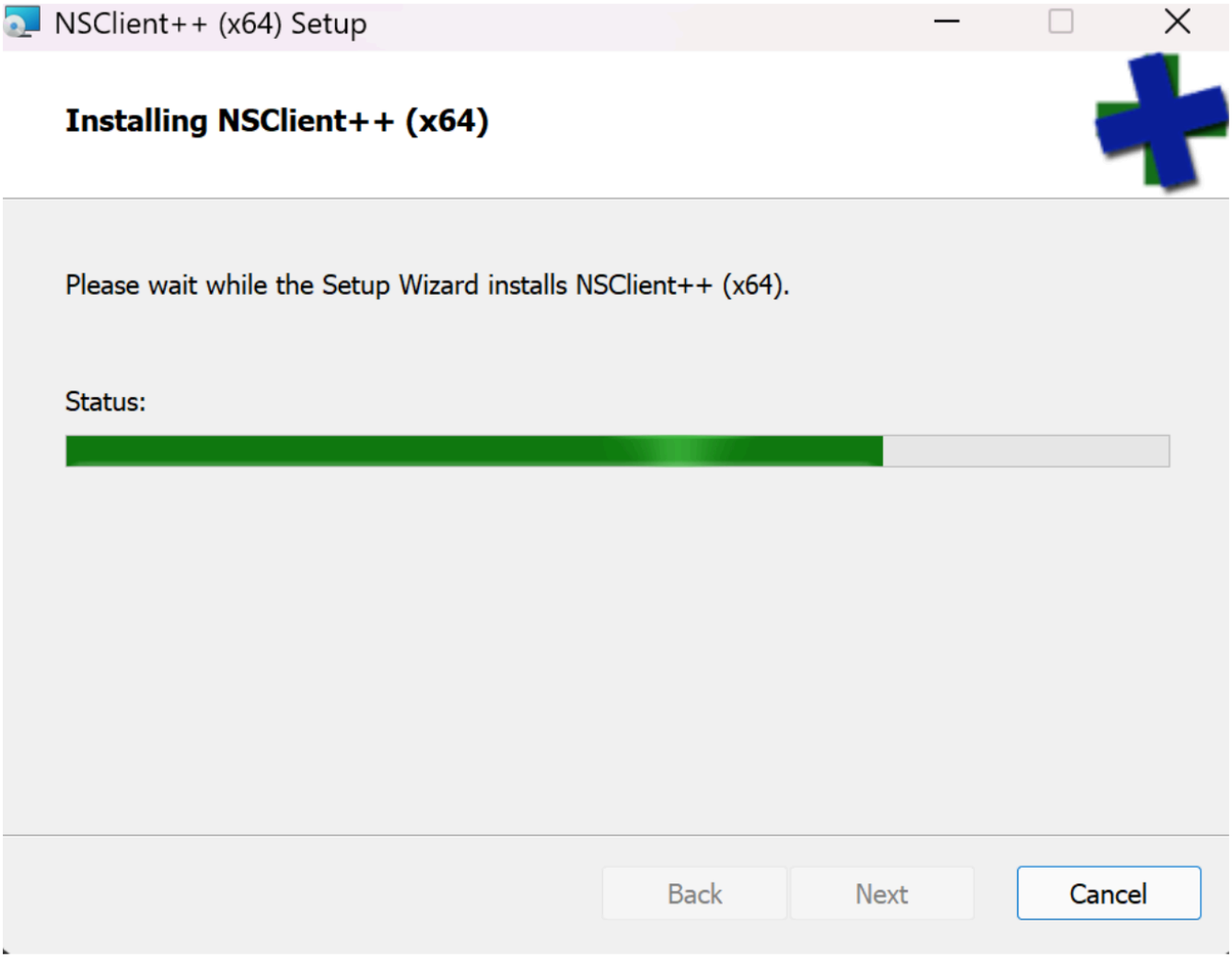
Reports

- Availability
- Trends
- Alerts
  - History
  - Summary
  - Histogram
- Notifications
- Event Log

System



Page Tour


NSClient++ (x64) Setup
Installing NSClient++ (x64)
Please wait while the Setup Wizard installs NSClient++ (x64).
Status:
Back
Next
Cancel

```
CheckHelpers = enabled

; Load on startup - If the module should be loaded on startup
CheckExternalScripts = enabled

; Load on startup - If the module should be loaded on startup
CheckNSCP = enabled


; in flight - TODO
[/settings/NRPE/server]

verify mode = peer-cert

tls version = tlsv1.2+

insecure = false


; in flight - TODO
[/settings/default]

allowed hosts = 127.0.0.1,192.168.29.181,172.17.16.1,172.17.0.0/16

password = vOiMR4smBNLkJBxs
```

```
PS C:\windows\system32> New-NetFirewallRule -DisplayName "Nagios NRPE" -Direction Inbound -LocalPort 5666 -Protocol TCP -Action Allow


Name                          : {494a7a4a-65cd-4bde-8907-5b4ddb608de0}
DisplayName                   : Nagios NRPE
Description                   :
DisplayGroup                  :
Group                         :
Enabled                       : True
Profile                       : Any
Platform                      : {}
Direction                     : Inbound
Action                        : Allow
EdgeTraversalPolicy           : Block
LooseSourceMapping            : False
LocalOnlyMapping              : False
Owner                         :
PrimaryStatus                 : OK
Status                        : The rule was parsed successfully from the store. (65536)
EnforcementStatus             : NotApplicable
PolicyStoreSource             : PersistentStore
PolicyStoreSourceType         : Local
RemoteDynamicKeywordAddresses : {}
PolicyAppId                   :
PackageFamilyName             :



PS C:\windows\system32> docker exec -it nagios4 /opt/nagios/libexec/check_nrpe -H host.docker.internal -n
I (0.19.0 2026-09-07) seem to be doing fine...

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug nagios4
    Learn more at https://docs.docker.com/go/debug-cli/
```

```
Nagios Core 4.5.12
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2026-03-25
License: GPL

Website: https://www.nagios.org
Reading configuration data...
   Read main config file okay...
   Read object config files okay...

Running pre-flight check on configuration data...

Checking objects...
        Checked 7 services.
        Checked 1 hosts.
        Checked 1 host groups.
        Checked 0 service groups.
        Checked 1 contacts.
        Checked 1 contact groups.
        Checked 23 commands.
        Checked 5 time periods.
        Checked 0 host escalations.
        Checked 0 service escalations.
Checking for circular paths...
        Checked 1 hosts
        Checked 0 service dependencies
        Checked 0 host dependencies
        Checked 5 timeperiods
Checking global event handlers...
Checking obsessive compulsive processor commands...
Checking misc settings...
```

```
Running pre-flight check on configuration data...

Checking objects...
        Checked 7 services.
        Checked 1 hosts.
        Checked 1 host groups.
        Checked 0 service groups.
        Checked 1 contacts.
        Checked 1 contact groups.
        Checked 23 commands.
        Checked 5 time periods.
        Checked 0 host escalations.
        Checked 0 service escalations.
Checking for circular paths...
        Checked 1 hosts
        Checked 0 service dependencies
        Checked 0 host dependencies
        Checked 5 timeperiods
Checking global event handlers...
Checking obsessive compulsive processor commands...
Checking misc settings...

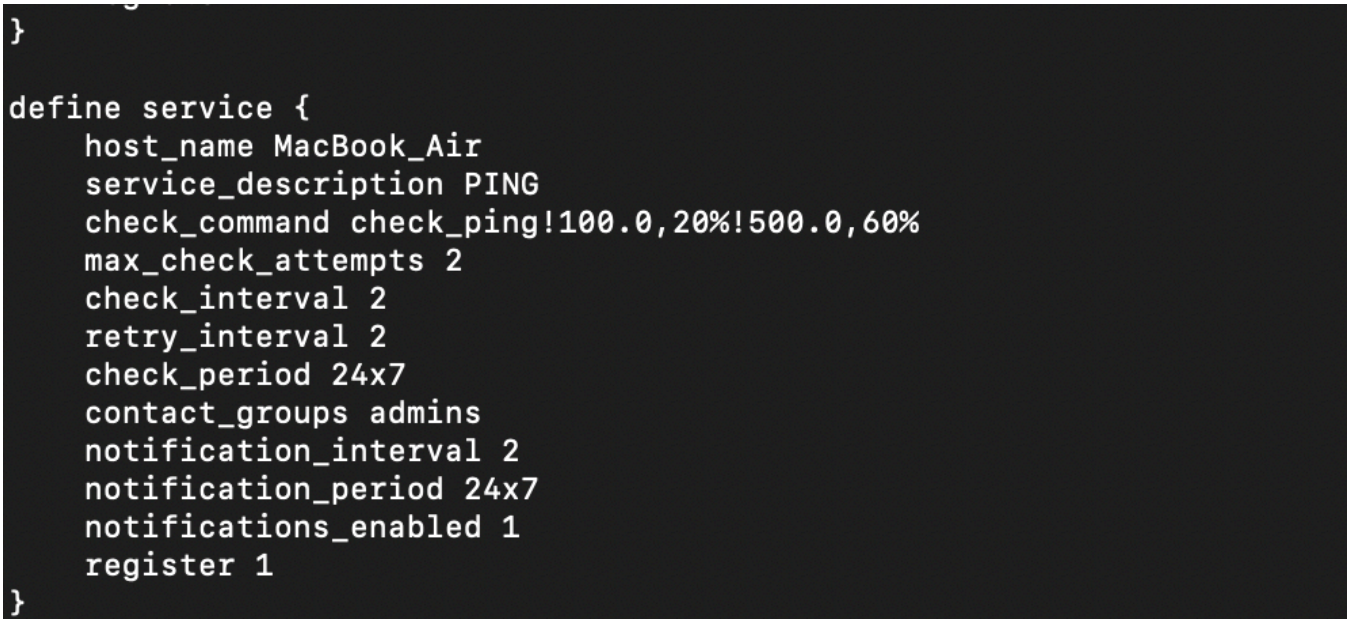
Total Warnings: 0
Total Errors:   0

Things look okay - No serious problems were detected during the pre-flight check

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug nagios4
    Learn more at https://docs.docker.com/go/debug-cli/
(base) irishadeshpande@Irishas-MacBook-Air ~ % docker exec nagios4 getent hosts host.docker.internal
192.168.65.254  host.docker.internal
(base) irishadeshpande@Irishas-MacBook-Air ~ %
```

```
(base) irishadeshpande@Irishas-MacBook-Air ~ % docker exec nagios4 mkdir -p /opt/nagios/etc/servers
(base) irishadeshpande@Irishas-MacBook-Air ~ % docker exec nagios4 sh -c 'cat > /opt/nagios/etc/servers/MacBook.cfg
<<EOF
define host {
    use generic-host
    host_name MacBook_Air
    alias MacBook Air
    address 192.168.65.254
    register 1
}

define service {
    host_name MacBook_Air
    service_description PING
    check_command check_ping!100.0,20%!500.0,60%
    max_check_attempts 2
    check_interval 2
    retry_interval 2
    check_period 24x7
    contact_groups admins
    notification_interval 2
    notification_period 24x7
    notifications_enabled 1
    register 1
}
EOF'
(base) irishadeshpande@Irishas-MacBook-Air ~ % docker exec nagios4 cat /opt/nagios/etc/servers/MacBook.cfg
define host {
    use generic-host
    host_name MacBook_Air
    alias MacBook Air
    address 192.168.65.254
    register 1
}
```



```
Nagios Core 4.5.12
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2026-03-25
License: GPL

Website: https://www.nagios.org
Reading configuration data...
   Read main config file okay...
   Read object config files okay...

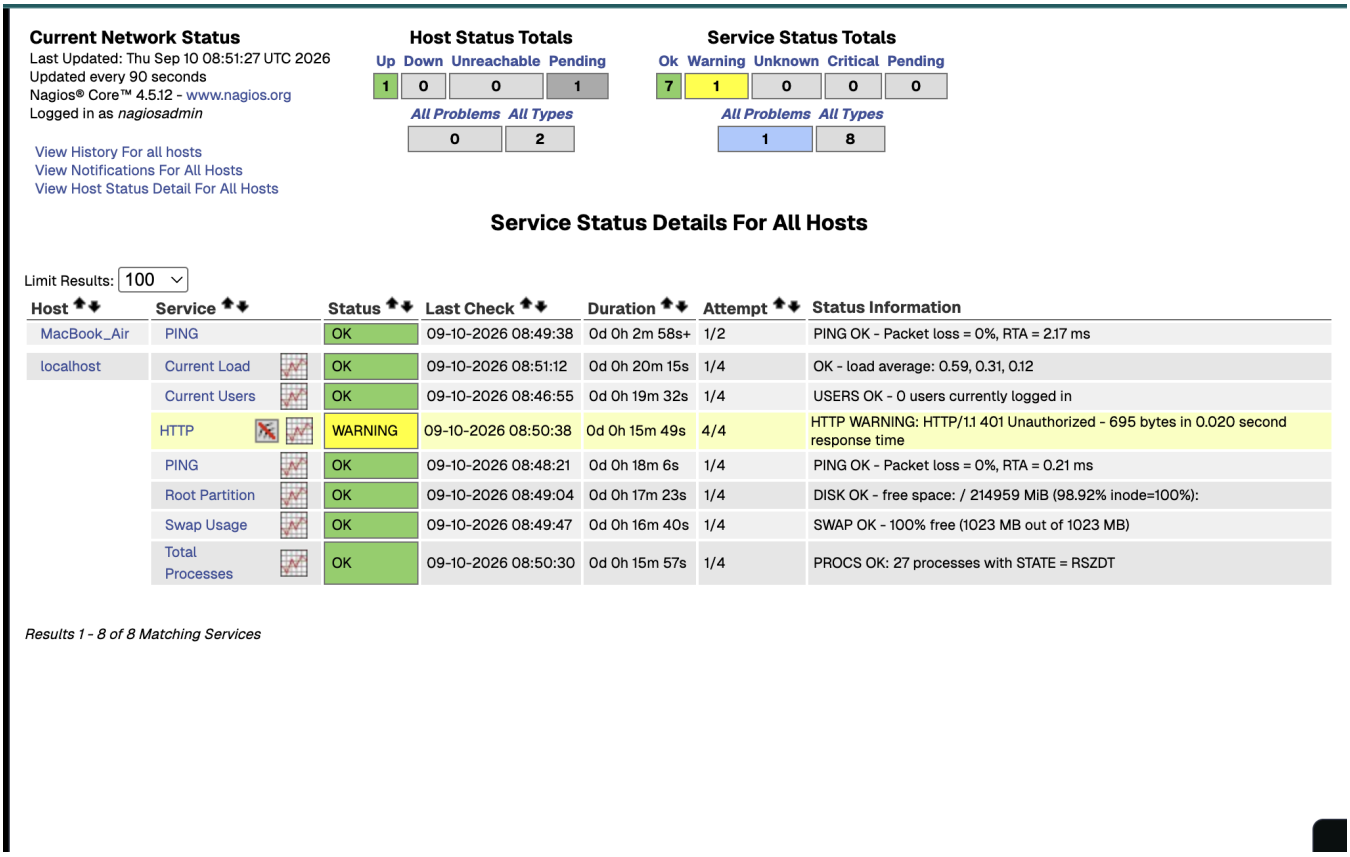
Running pre-flight check on configuration data...

Checking objects...
        Checked 7 services.
        Checked 1 hosts.
        Checked 1 host groups.
        Checked 0 service groups.
        Checked 1 contacts.
        Checked 1 contact groups.
        Checked 23 commands.
        Checked 5 time periods.
        Checked 0 host escalations.
        Checked 0 service escalations.
Checking for circular paths...
        Checked 1 hosts
        Checked 0 service dependencies
        Checked 0 host dependencies
        Checked 5 timeperiods
Checking global event handlers...
Checking obsessive compulsive processor commands...
Checking misc settings...

Total Warnings: 0
Total Errors:   0

Things look okay - No serious problems were detected during the pre-flight check
```

```
(base) irishadeshpande@Irishas-MacBook-Air ~ % docker exec nagios4 grep -nE 'cfg_(file|dir)' /opt/nagios/etc/nagios.cfg
29:cfg_file=/opt/nagios/etc/objects/commands.cfg
30:cfg_file=/opt/nagios/etc/objects/contacts.cfg
31:cfg_file=/opt/nagios/etc/objects/timeperiods.cfg
32:cfg_file=/opt/nagios/etc/objects/templates.cfg
35:cfg_file=/opt/nagios/etc/objects/localhost.cfg
38:#cfg_file=/opt/nagios/etc/objects/windows.cfg
41:#cfg_file=/opt/nagios/etc/objects/switch.cfg
44:#cfg_file=/opt/nagios/etc/objects/printer.cfg
48:# extension) in a particular directory by using the cfg_dir
51:#cfg_dir=/opt/nagios/etc/servers
52:#cfg_dir=/opt/nagios/etc/printers
53:#cfg_dir=/opt/nagios/etc/switches
54:#cfg_dir=/opt/nagios/etc/routers
76:# object configuration files (see the cfg_file and cfg_dir options above).
1347:cfg_dir=/opt/nagios/etc/conf.d
1348:cfg_dir=/opt/nagios/etc/monitor
```



Current Network Status
Last Updated: Thu Sep 10 08:51:27 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.12 - www.nagios.org
Logged in as *nagiosadmin*

View History For all hosts
View Notifications For All Hosts
View Host Status Detail For All Hosts

Host Status Totals

Up	Down	Unreachable	Pending
1	0	0	1

All Problems	All Types
0	2

Service Status Totals

Ok	Warning	Unknown	Critical	Pending
7	1	0	0	0

All Problems	All Types
1	8

Service Status Details For All Hosts

Limit Results: 100

Host	Service	Status	Last Check	Duration	Attempt	Status Information
MacBook_Air	PING	OK	09-10-2026 08:49:38	0d 0h 2m 58s+	1/2	PING OK - Packet loss = 0%, RTA = 2.17 ms
localhost	Current Load	OK	09-10-2026 08:51:12	0d 0h 20m 15s	1/4	OK - load average: 0.59, 0.31, 0.12
	Current Users	OK	09-10-2026 08:46:55	0d 0h 19m 32s	1/4	USERS OK - 0 users currently logged in
	HTTP	WARNING	09-10-2026 08:50:38	0d 0h 15m 49s	4/4	HTTP WARNING: HTTP/1.1 401 Unauthorized - 695 bytes in 0.020 second response time
	PING	OK	09-10-2026 08:48:21	0d 0h 18m 6s	1/4	PING OK - Packet loss = 0%, RTA = 0.21 ms
	Root Partition	OK	09-10-2026 08:49:04	0d 0h 17m 23s	1/4	DISK OK - free space: / 214959 MiB (98.92% inode=100%):
	Swap Usage	OK	09-10-2026 08:49:47	0d 0h 16m 40s	1/4	SWAP OK - 100% free (1023 MB out of 1023 MB)
	Total Processes	OK	09-10-2026 08:50:30	0d 0h 15m 57s	1/4	PROCS OK: 27 processes with STATE = RSZDT

Results 1 - 8 of 8 Matching Services

```
(base) irishadeshpande@Irishas-MacBook-Air ~ % docker exec nagios4 ping -c 4 192.168.65.254
PING 192.168.65.254 (192.168.65.254) 56(84) bytes of data.
64 bytes from 192.168.65.254: icmp_seq=1 ttl=63 time=0.414 ms
64 bytes from 192.168.65.254: icmp_seq=2 ttl=63 time=0.274 ms
64 bytes from 192.168.65.254: icmp_seq=3 ttl=63 time=0.289 ms
64 bytes from 192.168.65.254: icmp_seq=4 ttl=63 time=0.929 ms

--- 192.168.65.254 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3077ms
rtt min/avg/max/mdev = 0.274/0.476/0.929/0.266 ms
(base) irishadeshpande@Irishas-MacBook-Air ~ % docker exec nagios4 /opt/nagios/libexec/check_ping -H 192.168.65.254
-w 100.0,20% -c 500.0,60%
PING OK - Packet loss = 0%, RTA = 1.09 ms|rta=1.087000ms;100.000000;500.000000;0.000000 pl=0%;20;60;0
```

Conclusion:

Nagios was successfully implemented as a monitoring tool for monitoring hosts and services. The system was configured to perform continuous checks and display the status of monitored resources.Thus, Nagios provides effective monitoring and helps identify infrastructure and service issues.